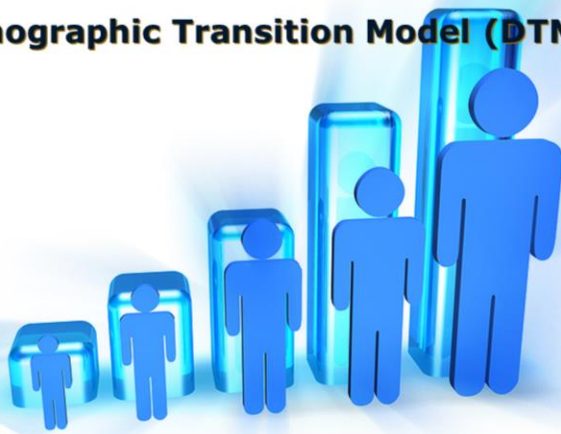


The Demographic Transition Model (DTM)



**RRHS APHG
2019-2020**



Key Demographic Vocabulary

- Crude Birth Rate
- Crude Death Rate
- Total Population
- Rate of Natural Increase (RNI)
- Population Doubling Time

You **MUST** know what these terms mean in order to make sense of the Demographic Transition Model.

Crude Birth Rate is the number of babies born per every 1,000 people in the population.

Crude Death Rate is the number of people who die per every 1,000 people in the population.

Total population is the sum of the entire population – it is generally an estimate based on a combination of different data, such as birth records and census projections.

The United States has a total population of 329,256,465 (July 2018 est.)

So, if the United States has a crude birth rate of 12.4, that means there were actually 4,082,780 babies born in 2018.

Also, the U.S. has a crude death rate of 8.2, so that means 2,699,903 people died in 2018.

So why don't we just look at the raw numbers? Because populations greatly vary in size. By looking at the number born/deaths per each 1,000 people in

the population, we are able to make comparisons. There were more babies born in the U.S. than there are the total amount of people in some countries, and as a result, we would not be able to take their actual number of births and compare them with the actual number of births in the U.S. without standardizing them in some way. Crude birth rate and crude death rates allow us to make those comparisons.

Same thing with the Rate of Natural Increase. Geographers and demographers use the crude birth and death rates to calculate the rate of natural increase because it is a standardized formula that allows for comparisons regardless of population size.

To calculate the **Rate of Natural Increase**, you subtract the crude death rate from the crude birth rate

$$\text{CBR} - \text{CDR} = \text{RNI}.$$

To put the statistic in percentage form, you then divide the RNI by 10.

the % of Natural increase (increase from births alone, not migration), it's $\text{CBR} - \text{CDR} = \text{RNI}/10$.

The RNI for the United States is $12.4 - 8.2 = 4.2/10 = .42\%$ So the population of the United States is growing by less than 1% each year through births alone.

Example #2: Zambia.

Total Population 16,445,079

CBR: 41.1

CDR: 12

There were a total of 675,893 births in Zambia last year (the raw number) – compare both the raw number and CBR to the U.S. Which one has a higher RNI? This is why we use Crude Birth Rate as a measure rather than the unstandardized raw data.

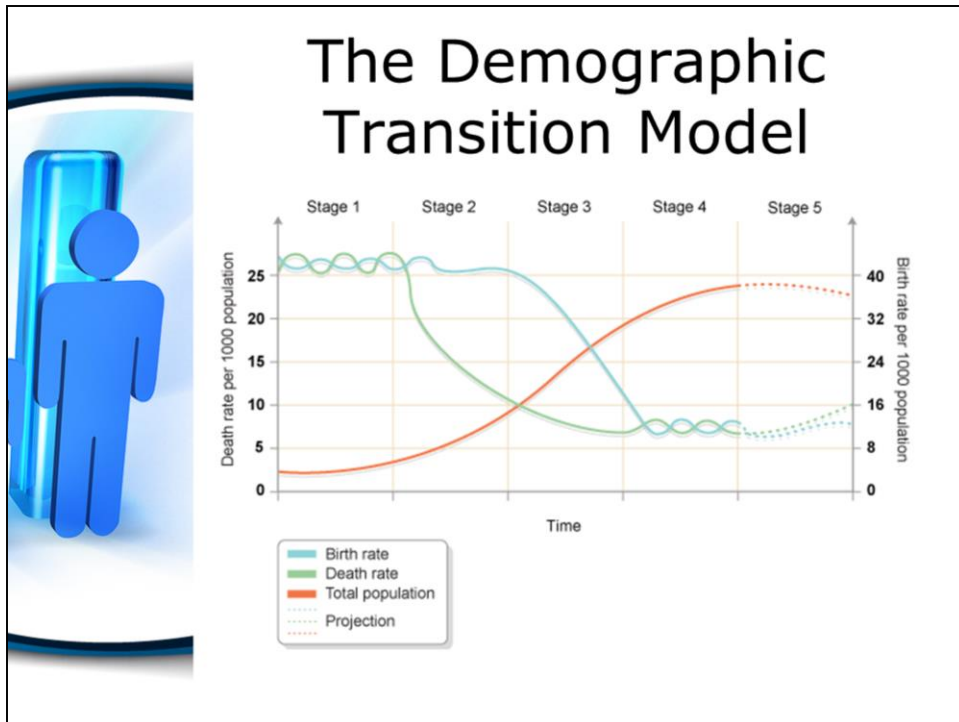
Zambia's RNI = $41.1 - 12 = 29.1/10 = 2.9$, so Zambia's population is growing by almost 3% each year. This is significantly faster growth than the U.S., even though the total number of children born in Zambia is drastically lower than the number born in the U.S.

Finally, to calculate the **Population Doubling Time** (the number of years it will take for the population to double), use the "Rule of 70", and divide 70 by the RNI. The "rule of 70" means that it will take 70 years for a population growing at 1% per year to double.

Doubling time for the U.S.: $70/.42 = 166$ years.

Doubling time for Zambia: $70/2.9 = 24$ years.

What needs to happen to accommodate twice as many people within a quarter of a century? Will the country be able to procure the resources for such a rapid expansion? What about services such as education and healthcare?



Why is Zambia's population growing so much faster than the U.S.'s? The rate can be explained with the country's level of economic development. The United States is a more developed country, and therefore experiences different demographic trends than countries of lower levels of development. A shift occurs in the demographics as a country shifts from one of lower economic development to one of more economic development. This shift can be visualized in the **Demographic Transition Model**.

The DTM is based on the experiences of countries that have already gone through the shift from less developed to newly industrialized to more developed – i.e., Western Europe and North America.

There are 5 stages of the Demographic Transition – we will look at each one and the characteristics of that stage individually.



Review of key demographics

- Life Expectancy
- Infant mortality rate
- Total fertility rate
- GNI per capita
- Access to sanitation
- Labor force by sector of the economy

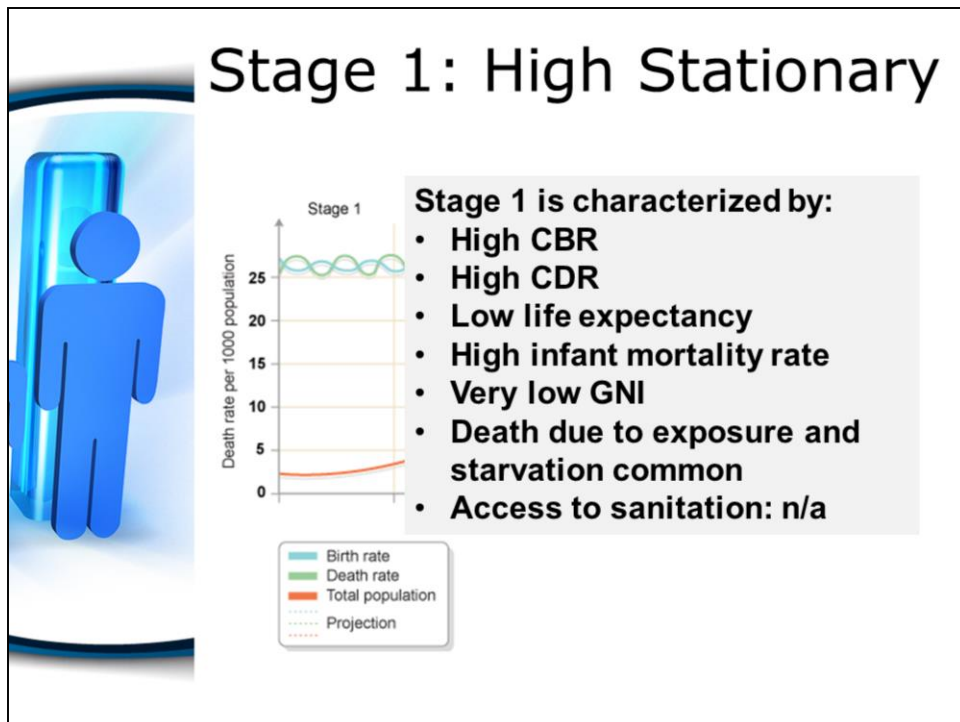
Life Expectancy: How many years a person born today can be expected to live.

Infant mortality rate: For every 1,000 babies born, the number that will die before their first birthday.

GNI per capita: Gross National Income per capita: The total sum of the value of all goods and services produced both within a country and their holdings abroad (outside of the country) divided by the total population. Similar to GDP per capita.

Access to sanitation: The percentage of people with operational plumbing within their homes.

Labor force by sector of the economy: The percentage of people who work in agriculture, manufacturing, or service jobs

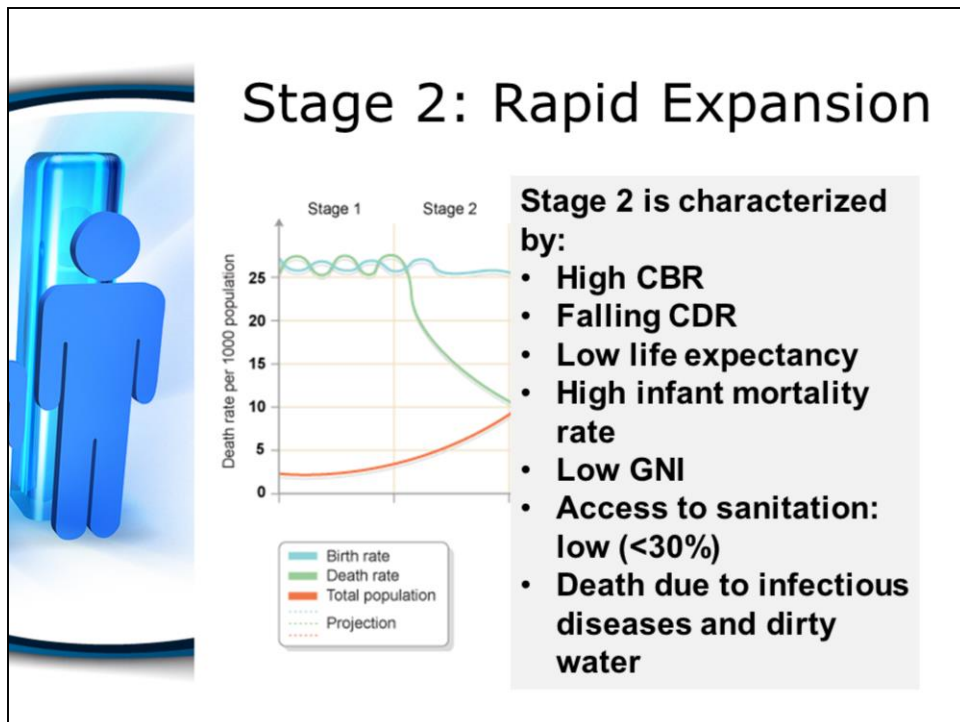


Stage 1 is called the “high stationary” stage – this means that both the CBR and CDR are high, but about equal, so the population is not growing because just as many people are dying as are being born. Both the birth rate and death rate have a great deal of fluctuation – during years of feast and abundant food, the birth rate will be high and the death rate low, but in years of famine (lack of food) the birth rate drops and the death rate soars.

Level of Economic Development? None – this stage is still characterized by nomadic hunter-gatherer tribes that are following food from place to place. They do not have a stable food supply.

As you learned in the Population Distribution Notes, there are only very few, isolated tribes that are nomadic hunter-gatherers today. Therefore, no *country* today is considered to be in Stage 1 of the DTM.

The RNI in stage 1 is low.



Stage 2 is characterized by a dramatic drop in the Crude Death Rate. What has happened that now so many fewer people are dying?

What is the major shift that has occurred? A significant shift occurs between stage 1 and stage 2 - the shift from nomadic hunter-gatherers to sedentary farmers in permanent civilizations. Countries in this stage have gone through it's Neolithic (agricultural) revolution. They now have a steady food supply, so deaths from malnourishment have drastically decreased. The Crude Birth Rate, however, is still high, though it is not fluctuating to the extent it was in Stage 1. Western Europe and North America stayed in this stage until about 1800.

Why is the CBR high? Families are still having many children because farming societies depend on the youth to help grow food for their families and also to take care of their parents in their old age, so it was not uncommon for families to have 8 or more children. This high total fertility rate (number of children born per woman) also helped to counter high infant mortality rates – if 10 children were born and only half of them survived to adulthood, then having more children acted as an “insurance” that at least a few would still be around to take care of the parents as they reached old age (also, note that often girls

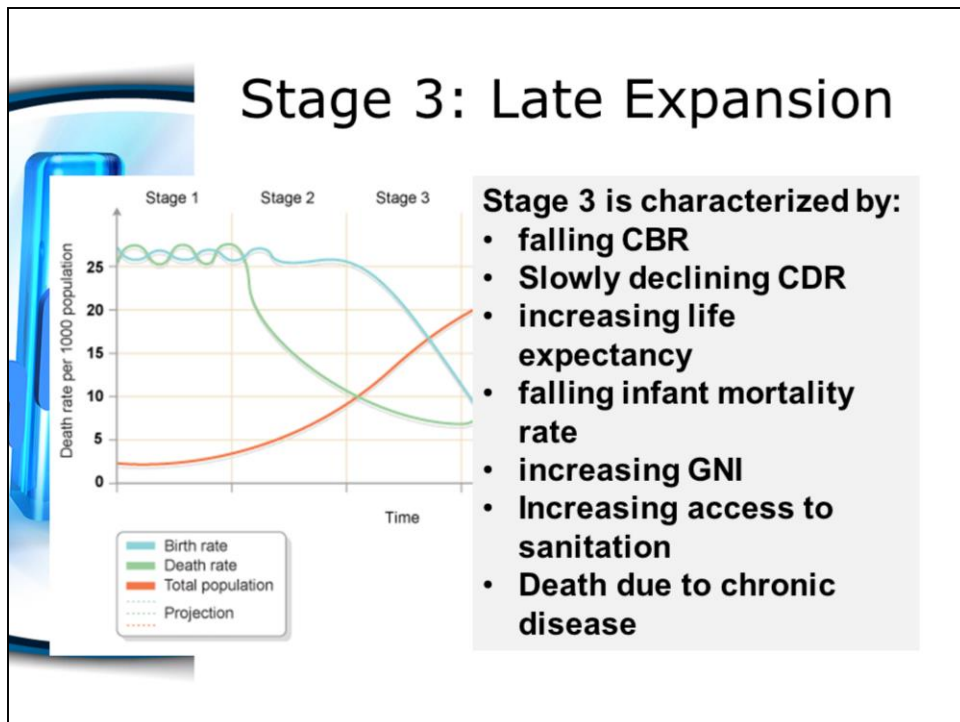
were married into other families and don't inherit property in many cultures, so there is preference for male children so the parents have an heir and provider).

Why has the CDR fallen? Since the advent of modern medicine (early 20th century/1900s), including antibiotics and vaccinations, the death rate in modern less developed countries has fallen even more drastically than they did in Western Europe and North America, who went through this stage prior to the discoveries in medicine (which occurred BECAUSE of the shifting demographics with increased economic development in those regions). However, due to a lack of access to clean water and improved sanitation, death rates are still higher than in later stages of development. Infectious diseases such as typhoid, malaria, cholera, dysentery, as well as vaccine-preventable diseases such as diphtheria, whooping cough, measles, and others are still a most-common cause of death. Malnutrition is common, though not as much as in stage 1, but enough to weaken the immune system of children who are then unable to fight off infectious diseases for which they have limited access to medicines because there is limited access to healthcare.

Why is the GNI low? The GNI is low because the vast majority of the area's population are engaged in farming as their economic activity. Subsistence farming (growing only enough food for one's family to consume) does not contribute much to the overall economy of a country (there is little to no money exchanged for the work being done). Countries in stage two often have a GNI per capita less than \$5,000. Most of these subsistence farming families live "off the grid", meaning they do not have access to public utilities such as electricity or water/sanitation services. Access to improved sanitation is low, especially in rural areas.

The RNI in Stage 2 countries is HIGH (typically >2%)

What stage of economic development is associated with this state on the DTM? Stage 2 is associated with Less Developed Countries.



Stage 3 is characterized by a dramatic drop in the Crude Birth Rate. Why are people having fewer children?

What is the major shift that has occurred? Countries in this stage of the DTM had experienced two major shifts – 1, they have gone through their industrial revolution and manufacturing now makes up a greater proportion of the GNI; and 2., the area now has more people living in cities than rural areas. More people living in cities means more people have access to education, especially for women, and increased access to healthcare, sanitation, and jobs outside of agriculture.

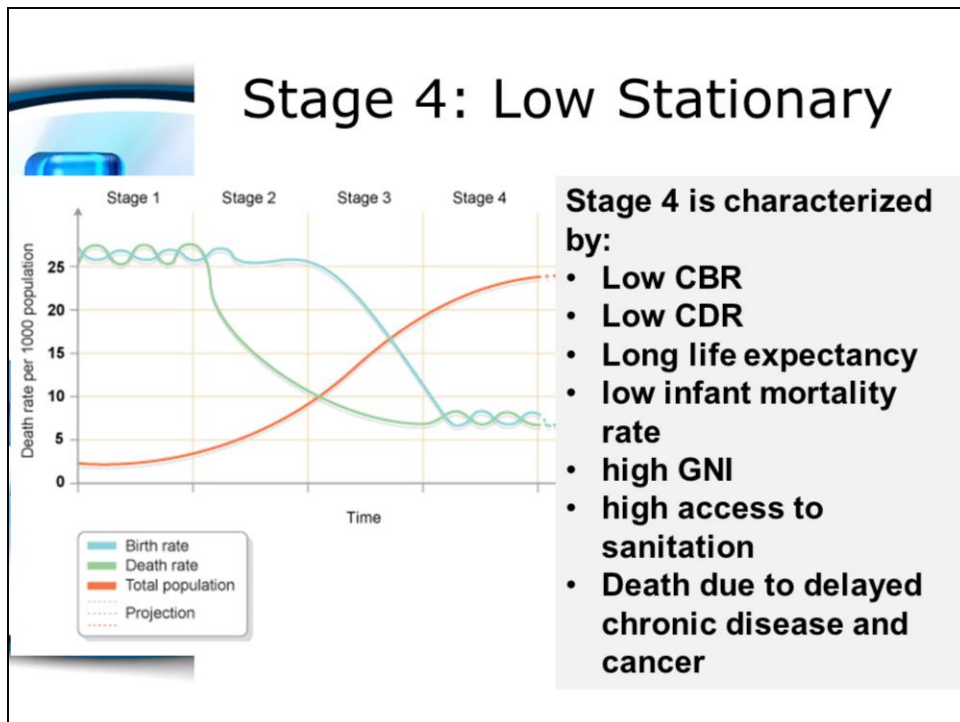
Why is the CBR falling? CBR is falling because education levels are increasing; women are staying in school longer and getting jobs which delays the age they marry and begin a family; women have greater access to healthcare; and infant mortality rates are falling because of improvements in sanitation, access to healthcare. Also, at this point, there is a shift in how children are viewed within the family – they are no longer additional helpers for food production, but are now an economic liability as parents must pay for their education, food (purchased instead of farmed), childcare while the parents work, etc.

What is happening to the CDR? The crude death rate continues to fall in this stage, just at a slower rate than in stage 2. Instead of dying from dirty water or infectious illnesses, now they are living longer and dying from chronic diseases such as cardiovascular disease (high blood pressure, coronary artery disease, etc.) and type 2 diabetes. These are treatable conditions for those with access to healthcare, and the life expectancy is now in the mid 60s.

Why is the GNI increasing? The GNI is increasing because more people are working skilled labor jobs that generate more income for the country – manufactured goods have a higher economic value than agricultural products, and are often exported which brings money into the country. GNI in stage 3 countries typically fall between \$8,000 and \$20,000 per capita.

RNI? The rate of natural increase is still high, but it is falling in this stage. Countries are still experiencing a significant population growth.

What stage of economic development is associated with this state on the DTM? Countries in this stage of economic development are generally classified as “newly industrialized”



Stage 4 is characterized by low birth rates and low death rates.

What is the major shift that has occurred? The major shift that has occurred in stage 4 is the shift to a service-based economy. As machinery increases, agricultural production and manufacturing becomes more efficient and fewer workers are needed to achieve similar output. Additionally, wages have become significantly greater, so production is often moved to other countries with lower wages. Education is universal, which means that high percentages of men and women finish secondary school (high school), and a large proportion continue on to trade school or college.

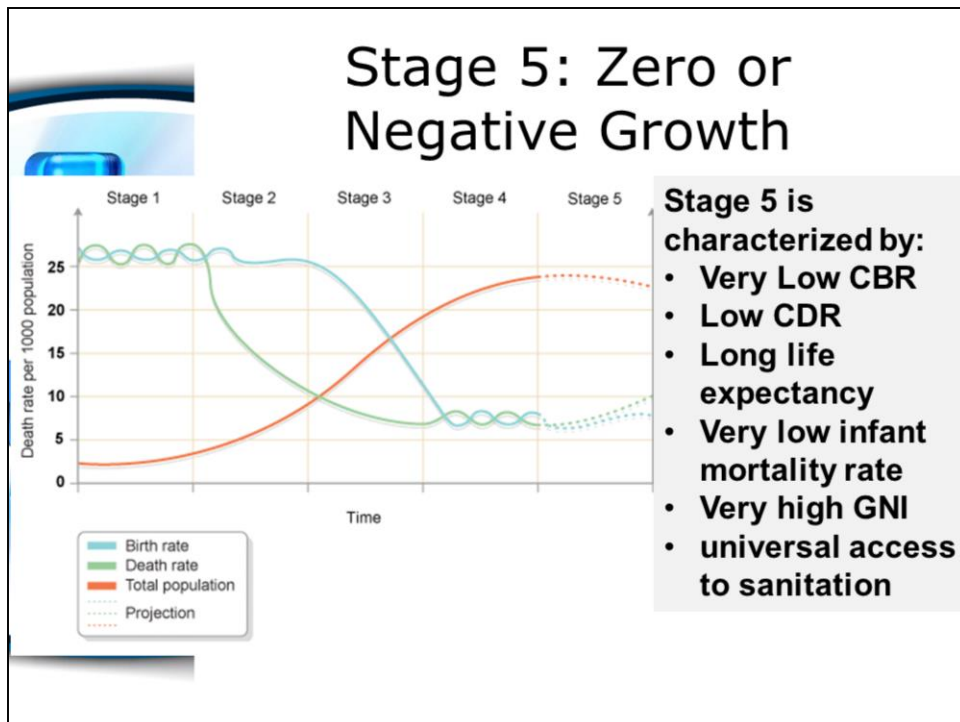
Why is the CBR low? Education levels are high, and many women pursue careers before starting families. At this stage, children are considered an economic liability – women often have careers outside the home and must pay for childcare and educational expenses. Social pressures regarding children and a woman's role in society have also changed. Access to healthcare is near-universal, and both men and women are educated on family planning (knowing how the human reproductive system works allows couples to plan when to have their children). Families often have just one or two children at this stage.

What is happening to the CDR? The CDR is low due to increased access to healthcare and near-universal access to improved sanitation. The cause of death in this stage is most likely delayed geriatric/chronic illnesses – those chronic illnesses present at stage 3 are delayed with medication and lifestyle changes, pushing the life expectancy to the late 70s/early 80s.

Why is the GNI increasing? The GNI per capita is increasing as skilled service work is a value-added economic contribution. Doctors, lawyers, teachers, computer programmers, researchers, finance, etc., are common service jobs.

RNI? The RNI has dropped significantly in Stage 4 countries – these countries are experiencing slow/stable growth. It is typically less than 1%.

What stage of economic development is associated with this state on the DTM? Stage 4 of the DTM is associated with More Developed Countries.



Stage 5 is characterized by birth rates falling below death rates – more people are dying than are being born. Stage 5 is a new addition to the DTM within the past 20 years or so. Since the model is based on the experience of Western Europe and North America, the model has been updated to reflect their continued demographic changes in recent decades. It is sometimes shown as “theoretical stage 5”.

What is the major shift that has occurred? The biggest shift that has occurred from stage 4 to stage 5 is that the birth rate has fallen below the death rate.

Why is the CBR falling? The CBR in stage 5 has continued to fall for the same reasons that it fell in stage 4 – women are highly educated, have careers, and children are an economic liability for parents. Marriage occurs later, and some couples decide not to have children at all. Life expectancy is very high – over 80 in many cases. Due to babies being born at a time in their parents’ lives where they are able financially to properly care for them, access to healthcare is high, and parents are knowledgeable about general health and nutrition, as well as infant safety such as sleeping in cribs, carseats, etc., and there are more public services available to those who cannot provide on their

own (welfare, subsidized childcare, etc.), infants have a much higher likelihood of surviving infancy and childhood. Children are more expensive in economies with

What is happening to the CDR? The CDR actually starts to increase slightly at this stage. This is because older people make up a larger percentage of the population as a whole, and therefore more of them will die per 1,000 people.

Why is the GNI so high? These countries are considered to be among the “MOST developed” countries – the vast majority of the population works in services, yet due to machinery and technology, their manufacturing output of high-valued factory-produced specialized goods is also high compared to the more labor-intensive manufacturing of Newly Industrialized Countries.

RNI? Since birth rates have fallen below death rates, the RNI in these countries is negative.

What stage of economic development is associated with this state on the DTM? Countries in stage 5 of the DTM are considered More Developed Countries. They are sometimes referred to as “Highly Developed Countries” to differentiate them from countries on the lower range of MDCs.

These countries have attained “Zero Population Growth” (Their RNI is Zero, and the population is not growing) or even negative growth, where the size of the population is actually declining. What are some challenges of a population that is declining in size where more people are dying than are being born?



The Epidemiological Transition Model

- Epidemiology = relating to the branch of medicine which deals with the incidence, distribution, and control of diseases.
- ETM looks at major causes of death related to stage of economic development.

Stage 1: (Stage 1/early stage 2 on DTM): Starvation, hunger, illness

Stage 2: (Stage 2/early stage 3 on DTM): communicable illnesses are the leading cause of death, particularly related to lack of access to improved sanitation (water-borne illness is a leading cause of infant mortality)

Stage 3 (Stage 3/early stage 4 on DTM): chronic illnesses such as type 2 diabetes or cardiovascular diseases such as coronary artery disease or high blood pressure

Stage 4 (Stage 4 through stage 5 of DTM): delayed chronic illnesses (treatments for type 2 diabetes and CVD are available)



SPEED Factors

- Social factors: Access to education, healthcare, sanitation
- Political factors: equal rights, stable government, trade agreements
- Economic factors: labor force by sector of the economy, GNI per capita
- Environmental factors – pollution, resources, climate and landforms
- Demographic factors – life expectancy, birth rates, death rates, and Infant mortality rate, RNI

Because stage of the DTM is tied to economic development, the SPEED factors of the DTM will be similar to those we use to discuss level of economic development.

Countries in stage 2 will have SPEED characteristics of Less Developed Countries

Countries in stage 3 will have SPEED characteristics of NICs

Countries in stage 4 and 5 will have SPEED characteristics of MDCs



Demographic Factors

- Fertility rates
 - Crude birth rate
 - Total Fertility Rate
 - Replacement rate
- Mortality rates
 - Crude Death Rate
 - Infant mortality rate
- Migration
 - In-migration (immigration)
 - Out-migration (emigration)

For Demographic factors, for population growth, geographers look more specifically at three main demographic factors: Fertility, mortality, and migration.

Fertility means birth – how many babies are being born. There are two main measures of fertility – **crude birth rate** tells us how many babies are born per 1,000 members of the society. **Total fertility rate** tells us how many babies are born *per woman* over her entire life. Both rates are important for understanding, but total fertility rate is important in that it reveals if a country's fertility is above or below **replacement rate**.

Replacement rate is a constant rate of 2.1. This is what the TFR MUST be in order for the population to maintain its current size (neither growing nor declining). Why? 2 children are necessary to replace the mother and her partner. However, because infant mortality will never be zero due to genetic issues, childhood cancers, and accidents; and that some people are unwilling or unable to have children at all, the TFR must be slightly above 2.

Mortality means death – Geographers look at two key statistics about death: the **crude death rate**, which gives an idea of the overall health of a country,

and also the infant mortality rate. Geographers can infer a great deal of information from an area's **infant mortality rate** because social factors such as access to sanitation, access to healthcare, and education for women all have a significant influence on the number of infants who die before their first birthday.

Migration is looking at people who are moving from one place to another, and a population will grow if more people are moving in than are moving out.



The Demographic Equation

- RNI does not include migration
- Demographic equation is a measure of population growth that **does** include migration

$DE = ((CBD - CDR) + (in-migration - out-migration)) / \text{total population}$, and can often explain



Notable Exceptions to the DTM

- Countries that don't fit the model
 - Russia's population sink
 - Tourism economies
 - Countries with high rates of HIV

Like all models, there are exceptions.



Russia's Population Sink

- Legacy of communist gov't and command economy
- Deteriorating social conditions in 1980s and 1990s
- Environmental damage
- Weak economy and lack of optimism for future

Economic development under the Communist government of the Soviet Union (USSR) was rampant. Command economies, like the USSR until its collapse, did not follow market forces, rather they were designed by the central government. The government controlled all economic activity – from farming to manufacturing to services. The result was a very weak economy – and extreme environmental damage. Unlike a market-driven economy (aka, free enterprise, capitalist, etc.) where the goal of privately-held firms (companies) is to maximize *profit* by minimizing costs, in a command economy the goal is to maximize *production* at minimal cost. You may think these sound like the same thing, but they are not.

This is a very simplified explanation with significant generalization, but for purposes of understanding...

In a free-market society, if a firm cuts production costs to the point of damaging quality of the product or their resources, people will no longer purchase their product and demand will drop, or they will lose their ability to maintain production because of the damage, and the company will lose profits. Therefore, their ultimate goal is to keep the quality at the level the consumers equate with the price, thereby maximizing their profits by having a reputable brand. If the environmental or social conditions of the factory are seen as unfavorable by the consumers, there may be pushback for for better working

conditions or more environmentally friendly practices. If a firm cannot keep workers and has to constantly replace and retrain them, their production will take a hit. Command economies do not have these market “checks” – employees are assigned, not free-will. Prices are set by the government. Production rates are set by the government. The ultimate goal is to meet those production goals at the lowest cost, and the government decides how to do that. The results have a lasting legacy, as is seen by the case of the former USSR/Russia today.

Russia’s economic development stagnated at the Newly Industrialized phase. Socially, politically, economically, and environmentally that is where they remain – stage 3. Demographically, however, they are in a population “sink” – birth rates are dramatically low, but for reasons that are different than countries in Stage 5 (mostly in Western Europe, and also Japan). The Life Expectancy is low, especially for men, and can be attributed to high rates of untreated cardiovascular disease, accidents, murder, suicide, and alcoholism, particularly among men. In the 1990s, life expectancy for men fell from 65 to 57 years. Lack of quality healthcare during the final decades of the Soviet Union and the decade after its collapse (1980s-2000s) are a main contributor to this trend. Infant mortality rates and instances of birth defects due to vitamin D deficiencies were increasing, and an estimated 75% of Russian women experienced complications during pregnancy and delivery that could have been prevented with appropriate prenatal medical care.

Combined with a weakened economy as global production has shifted to East Asia, Southeast Asia, and South America, the economic collapse that ultimately contributed to the collapse of the USSR, and a shift as post-USSR Russia moves more toward a mixed economic system (some government owned, some privately held), birth rates in the country have plummeted to among the lowest in the world.

Environmentally, the land suffered extreme damage due to soviet industrial development as few environmental safeguards were put into place (they are expensive) – up to 15% of Russia’s land is contaminated. 75% of the rivers are unsuitable for drinking water due to pollution, and 50% of tap water is unsafe to drink due to insufficient infrastructure. One of the biggest environmental disasters of the U.S.S.R. was the meltdown at the Chernobyl nuclear power plant – which went unreported to the rest of the world until radiation sensors in other countries started picking up the nuclear fallout. Chernobyl is in modern day Ukraine.

[World Watch](#). 1996 Jan-Feb;9(1):22-3.

Russia's population sink.

[Nelson T.](#)



Tourism Economies

- Does not fit DTM progression
- Uneven social development in country
- Resources diverted from locals to tourists
- Profits often leave the country



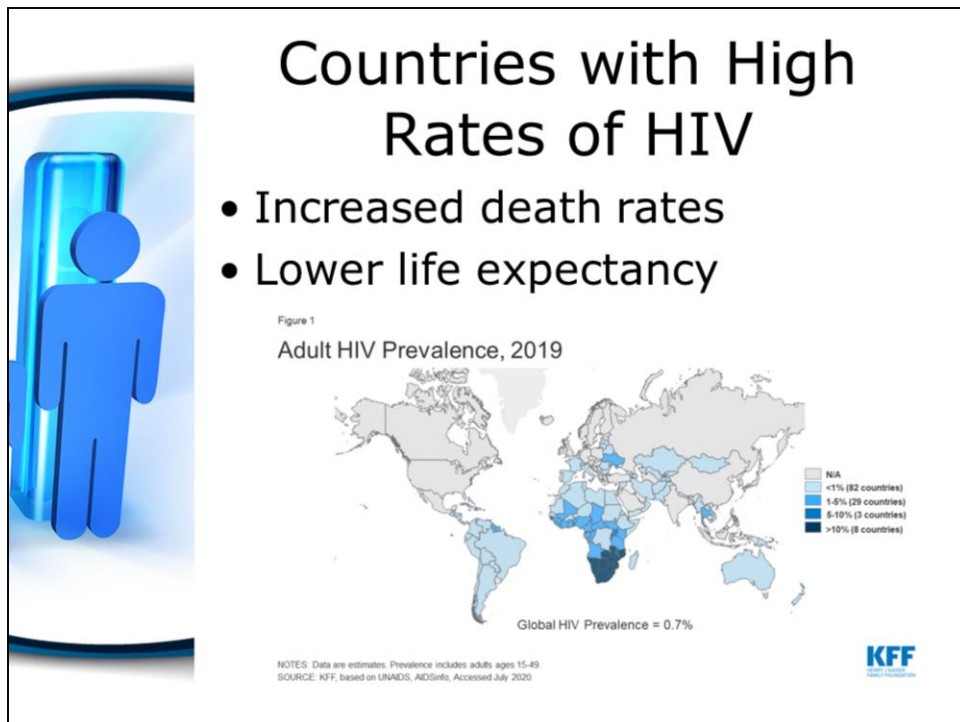
Tourism can have many economic benefits – jobs for locals, an influx of cash from people outside of the area (they come spend their money in local businesses! Yay!), and a desire to make a destination attractive so others will want to come see it. Many countries, particularly smaller countries with desirable physical geographic features such as clear ocean waters and nice beaches, or mountains for climbing or skiing, hope to increase tourism as an economic base for their society.

These so-called “tourism” economies often do not follow the path of Western Europe and North America on their way to development. There is often a jump from primary economic activity (agriculture) straight to services, with the resulting increase in GNI per capita.

However, social development is very uneven – infrastructure is often built for resorts, but do not reach into the surrounding communities. Resources (water, food, luxuries) are diverted for the tourist, while the citizens may be poor. Foreigners are often brought in to work “front of house” jobs (such as front desk, tour guides, etc.) which are higher paying than “back of house” (kitchen, janitorial, maintenance) jobs. Resorts are often also owned by foreign corporations, which means the profits do not go back into the local economy to

help its development. What this means is that while some parts of these tourism countries are thriving and may seem to be well developed, other parts are still stuck in stage 2 of the DTM and exhibit the characteristics of LDCs.

The image above shows this divide – on the left side of the picture are the beautiful beaches and lush green of a luxury resort in Zanzibar (a tourist destination in Tanzania on the Indian Ocean), while the right side of the image shows the slums that house the locals. The only thing separating them is the fence around the resort, but there is significant inequality of social development.



HIV, the virus that causes AIDS (acquired immunodeficiency syndrome), is one of the world's most serious health and development challenges. Approximately 38 million people are currently living with HIV, and tens of millions of people have died of AIDS-related causes since the beginning of the epidemic.

As you can see on the map above, the region of Southern Africa has the highest Adult HIV rate of any region of the world – more than 10% of the adult population in this region has the virus that causes AIDS. Many do not have access to treatments that slow the progression of the virus, such as antiretroviral drugs.

HIV/AIDS drastically lowers the life expectancy of those affected by weakening their immune systems – this means that bacterial infections and viruses a person with a healthy immune system could easily fight off, such as a the common cold, or strep throat, or a sinus infection, can become deadly to those with and AIDS-weakened immune system.

The impact of HIV/AIDS is that the death rate increases earlier in the DTM – For example – South Africa, a country in Stage 3, has exhibited an increasing death rate, which would not normally occur until the latter parts of stage 4 or

stage 5.